

Statistical Mechanics of Money:

Applying Boltzmann-Gibbs Distributions and Entropy
to Economic Dynamics via Agent-Based Simulation

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Abstract

In both economics and physics, the limiting behavior for superposed effects of large numbers of individuals (particles) is due to probabilistic central limit theorems[1]. Economic models leading to non-Gaussian distributions often involve complex or heterogeneous interactions among market participants. There are several references with statistical mechanics flavor in economic modeling, which describe equilibria via maximum entropy configurations subject to constraints such as fixed total wealth.

Keywords: Statistical mechanics, Boltzmann-Gibbs distribution, Gibbs entropy, Agent-based simulation, Markov processes, Economic dynamics, Complex systems.

1 Introduction

The applicability of **statistical mechanics** to economic systems, specifically how money distribution in a society of interacting economic agents follows the **Boltzmann-Gibbs distribution**. [2, 3] This is a foundational principle from thermodynamics and statistical physics. The assumption is that money, like energy in a physical system, is conserved during transactions between agents.

I conducted agent based simulations modeling economic transactions as random exchanges of money between agents, analogous to particle collisions in a gas. Under various transaction mechanisms such as constant exchanges between agents, fraction γ of individual wealth per agent in each transaction, fraction of system average, etc., the simulated money distribution converges to an exponential form $P(m) = \frac{1}{T}e^{-m/T}$, where $T = M/N$ represents the system's effective "temperature" (average wealth per agent).

Two simulation models are analyzed:

1. **Basic money distribution** with no debt constraints
2. **System with debt** allowing limited negative balances

It's possible to demonstrate that **Gibbs entropy** correctly predicts equilibrium states and that the system evolves as a **Markov process** toward maximum entropy. These results validate the hypothesis that economic inequality and wealth concentration emerge as natural equilibrium states when agents interact via random transactions, analogously to how physical systems spontaneously approach thermodynamic equilibrium.

This research connects fundamental concepts from statistical mechanics from Markov processes, Boltzmann-Gibbs distributions, Gibbs entropy, and dynamical systems theory to economics and the so called "econophysics" or "social physics", providing a computational validation of theoretical predictions.

1.1 Motivation

The term “econophysics” was introduced by analogy with similar terms, such as astrophysics, geophysics, and biophysics, which describe applications of physics to different fields.

It should be emphasized that econophysics does not literally apply the laws of physics, such as Newton’s laws or quantum mechanics, to humans, but rather uses mathematical methods developed in statistical physics to study properties of complex economic systems consisting of a large number of humans. So, it may be considered as a branch of applied theory of probabilities. Econophysics distances itself from the verbose, narrative, and ideological style of political economy and is closer to econometrics in its focus. Studying mathematical models of a large number of interacting economic agents, econophysics has much common ground with the agent-based modeling and simulation [4].

1.2 Boltzmann-Gibbs Distribution of energy

The fundamental law of equilibrium statistical mechanics is the Boltzmann–Gibbs distribution. It states that the probability $P(\varepsilon)$ of finding a physical system or subsystem in a state with the energy ε is given by the exponential function

$$P(\varepsilon) = ce^{-\varepsilon/T}, \quad (1)$$

where T is the temperature, and c is a normalizing constant. Here we set the Boltzmann constant k_B to unity by choosing the energy units for measuring the physical temperature T . Then, the expectation value of any physical variable x can be obtained as

$$\langle x \rangle = \frac{\sum_k x_k e^{-\varepsilon_k/T}}{\sum_k e^{-\varepsilon_k/T}}, \quad (2)$$

where the sum is taken over all states of the system. Temperature is equal to the average energy per particle: $T \sim \langle \varepsilon \rangle$, up to a numerical coefficient of the order of 1.

Equation (1) can be derived in different way. One of the shortest derivations can be summarized as

follows. Let us divide the system into two (generally unequal) parts. Then, the total energy is the sum of the parts: $\varepsilon = \varepsilon_1 + \varepsilon_2$, whereas the probability is the product of probabilities: $P(\varepsilon) = P(\varepsilon_1)P(\varepsilon_2)$. The only solution of these two equations is the exponential function (1).

A more sophisticated derivation, proposed by Boltzmann himself, uses the concept of **entropy**. Let us consider N particles with the total energy E . Let us divide the energy axis into small intervals (bins) of width $\Delta\varepsilon$ and count the number of particles N_k having the energies from ε_k to $\varepsilon_k + \Delta\varepsilon$. The ratio $N_k/N = P_k$ gives the probability for a particle to have the energy ε_k . Let us now calculate the **multiplicity** W , which is the number of permutations of the particles between different energy bins such that the occupation numbers of the bins do not change. This quantity is given by the combinatorial formula in terms of the factorials

$$W = \frac{N!}{N_1!N_2!N_3!\dots}. \quad (3)$$

The logarithm of multiplicity is called the entropy $S = \ln W$. In the limit of large numbers, the entropy per particle can be written in the following form using the Stirling approximation for the factorials

$$\frac{S}{N} = - \sum_k \frac{N_k}{N} \ln \left(\frac{N_k}{N} \right) = - \sum_k P_k \ln P_k. \quad (4)$$

Now we would like to find what distribution of particles between different energy states has the highest entropy, i.e., the highest multiplicity, provided that the total energy of the system,

$$E = \sum_k N_k \varepsilon_k, \quad (5)$$

has a fixed value. Solution of this problem can be easily obtained using the method of Lagrange multipliers [5], and the answer gives the exponential distribution (1).

The same result can be derived from the **ergodic theory**, which says that the many-body system occupies all possible states of a given total energy with equal probabilities. Then it is straightforward to show [5] that the probability distribution of the

energy of an individual particle is given by Eq. (1).

2 Model and Theory

As previously discussed, we can assume that agents can be modeled as particles in a closed system, where money is conserved. Instead of exchanging energy, agents exchange money through random transactions. The system evolves through random transactions, forming a Markov process. The theoretical framework predicts that the wealth distribution converges to the Boltzmann-Gibbs distribution, with entropy increasing over time until equilibrium is reached.

2.1 Conservation of Money

One could assume that Boltzmann's derivation of the energy distribution can be applied to other statistical systems with a conserved quantity, such as money. Indeed, we can consider the economy as a closed system with a fixed number of agents N and a total amount of money M . Therefore, the total money in the system is conserved. As a starting point, we can consider a simple model where agents exchange money through random transactions, and the total amount of money remains constant.

Introducing the concept of money conservation we can assume the model **without debt**, where agents cannot have negative balances. So the conditions for this model are:

- $\sum_{i=1}^N m_i = M$.
- $m_i \geq 0$, for $i = 1, \dots, N$.
- $\delta m_i = -\delta m_j$, for $i \neq j$.
- $\delta m < m_i$.

2.2 Boltzmann-Gibbs Distribution of Money

As argued in previous section, we can associate the money distribution to the energy exchange in a physical system. For this reason, we can assume that the probability of finding an agent with money

m is given by the Boltzmann-Gibbs distribution:

$$P(m) = ce^{-m/T}, \quad (6)$$

where c is a normalization constant and T is the effective temperature of the system: $T = M/N$, which is average money per agent.

The **entropy** of the system is defined as:

$$S = - \sum_{i=1}^N P(m_i) \log P(m_i). \quad (7)$$

The **equilibrium** is reached when the entropy is maximized when the system "cools off" to the Boltzmann-Gibbs distribution.

2.2.1 Transaction Mechanisms

Many different kinds of transactions can be considered, such as:

- **Constant transactions:** $\Delta m = 1$ where a fixed amount of money is exchanged between agents.
- **Fraction of Pair's Average:** A random fraction of the average money of the pair $\frac{m_i + m_j}{2}$ is exchanged: $\Delta m = r \cdot \frac{m_i + m_j}{2}$,
- **Fractional transactions** where a fraction γ of the agent's money is exchanged, the two agents will exchange only a portion of their money. This mechanism show how richer agents can exchange more money than poorer agents, and it is more realistic.

Therefore we can reassume the transaction mechanisms as:

$$\Delta m = \begin{cases} 1, & \text{constant exchange,} \\ \eta \frac{x_i + x_j}{2}, & \text{fraction of pair average,} \\ \eta \bar{x}, & \text{fraction of system average,} \end{cases} \quad 0 \leq \eta \leq 1.$$

Obviously, the transaction is accepted only if the payer has sufficient funds in the chosen pool.

2.3 The different models

Different assumptions were made in the literature to model the money exchange between agents.

The aim it's to find a model that can reproduce the empirical wealth distribution observed in real economies.

2.3.1 Model without debt

The simplest model has constant transactions and no debt can be accumulated. This model leads to the Boltzmann-Gibbs distribution as previously discussed.

2.3.2 Model with debt

The exact mechanisms of limiting debt in the real economy are complicated and obscured. In this scenario, debt is introduced in the model [6]. Debt may be considered as negative money. When an agent borrows money from a bank (considered here as a big reservoir of money), the cash balance of the agent (positive money) increases, but the agent also acquires a debt obligation (negative money), so the total balance (net worth) of the agent remains the same, and the conservation law of total money (positive and negative) is satisfied. Any stable economic system must have a mechanism preventing unlimited borrowing and unlimited debt. Otherwise, agents can buy any products without producing anything in exchange by simply going into unlimited debt.

A simple model was considered where the maximal debt of any agent is limited by a certain amount m_d . This means that the boundary condition $m_i \geq 0$ is now replaced by the condition $m_i \geq -m_d$ for all agents i . Not surprisingly, the stationary money distribution again has the exponential shape, but now with the boundary condition at $m = -m_d$ and the higher money temperature

$$T_d = m_d + \frac{M}{N}. \quad (8)$$

By allowing agents to go into debt up to m_d , we effectively increase the amount of money available to each agent by m_d . So, the money temperature, which is equal to the average amount of effectively available money per agent, increases. Therefore, we see that debt does not violate the conservation law of money, but rather modifies boundary conditions for $P(m)$.

One way of limiting the total debt in the economy is the so-called required reserve ratio r [7]. Every bank is required by law to set aside a fraction r of the money deposited into the bank, and this reserved money cannot be loaned further. If the initial amount of money in the system (the monetary base) is M_0 , then with loans and borrowing the total amount of positive money available to agents is

$$M = \frac{M_0}{r}, \quad (9)$$

where the factor $1/r$ is called the money multiplier. This is how banks "create money" in the modern economy. They do so by creating debt obligations in the system, where the maximal debt (limited by the factor $1/r$) can be:

$$M_d = M - M_0 = \frac{M_0}{r} - M_0 = M_0 \frac{1-r}{r}. \quad (10)$$

When the debt is maximized, the money distribution is again exponential but now "double sided". Both total amounts of positive $\frac{M_0}{r}$ and negative $\frac{M_0(1-r)}{r}$ money are conserved separately since money circulate between the agents in the system, so there are effectively two conservation laws for each of them.

Let $r \in (0, 1)$ denote the fraction of agents in the positive sector and $1 - r$ the fraction in the debt sector. The populations are initialized as

$$N_+ = rN, \quad N_- = (1-r)N,$$

and the corresponding total amounts of money are

$$M_+ = \frac{M_0}{r}, \quad M_- = \frac{M_0(1-r)}{r}.$$

This choice ensures that the total net wealth remains $M_+ - M_- = M_0$. Hence the system has a positive money sector and a negative debt sector, while preserving the net conserved amount M_0 .

The two pools evolve independently, and at the end the debt pool is reflected to negative values:

$$(x_1, \dots, x_{N_+}, -y_1, \dots, -y_{N_-}).$$

This construction produces a bimodal distribution with a positive branch and a negative branch, separated by a zero-crossing at $m = 0$. The pa-

parameter r controls the relative weight of the two sectors. To determine the value of the reverse ratio that aligns the two branches at the origin, we impose the matching condition

$$p_+(0) = p_-(0),$$

where the two densities are

$$P(m_+) = \frac{N_+}{N} \frac{1}{T_+} e^{-m_+/T_+}, \quad m_+ \geq 0,$$

$$P(m_-) = \frac{N_-}{N} \frac{1}{T_-} e^{m_-/T_-}, \quad m_- \leq 0.$$

2.3.3 Proportional transfers

A different model [8] was introduced as a concept of saving propensity, it has the name of multiplicative asset exchange model. This model perfectly fits the money conservation, thus it considers that agents will exchange only a fraction of their money, and the rest will be saved. To make it simple, each agent exchanges a fraction γ of their money, where γ is a fixed number between 0 and 1:

$$\Delta m = \gamma m_i. \quad (11)$$

In this model, the stationary distribution of money is not the Boltzmann-Gibbs distribution, but a Gamma distribution:

$$P(m) = cm^\beta e^{-m/T}, \quad (12)$$

where it differs from the exponential distribution (Eq. 6) by the power law factor m^β , where β is a parameter that depends on the saving propensity γ . From Boltzmann-Gibbs kinetic theory that is discussed in 2.4, [8] derived the relation between β and γ :

$$\beta = -1 - \ln(2)/\ln(1 - \gamma). \quad (13)$$

which shows that if payers spend a relatively small fraction of their money, $\gamma < 1/2$, then the low money population is suppressed $P(m=0) \rightarrow 0$.

This transactional model has a difference with respect to the previous ones: the transactions are not reversible, indeed when an agent i gives a fixed fraction γ of his money m_i to an agent with balance

m_j , their balances become $(1-\gamma)m_i$ and $m_j + \gamma m_i$. If we try to reverse this process and appoint the agent j to be the payer and to give the fraction γ of her money, $\gamma(m_j + \gamma m_i)$, to the agent i , the system does not return to the original configuration $[m_i, m_j]$. This is better discussed in the next section (2.4), where the non-equilibrium dynamics of the system is analyzed.

2.4 Non-equilibrium dynamics

We can study the evolution of the system that is not in equilibrium, and we can analyze the time-dependent behavior of the system starting from the Boltzmann kinetic equation [9]:

$$\begin{aligned} \frac{\partial P(m, t)}{\partial t} = & \int_0^\infty \int_0^\infty \left[-f_{m, m' \rightarrow m-\Delta, m'+\Delta} P(m) P(m') + \right. \\ & \left. + f_{m-\Delta, m'+\Delta \rightarrow m, m'} P(m-\Delta) P(m'+\Delta) \right] dm' d\Delta. \end{aligned} \quad (14)$$

Here $f_{[m, m'] \rightarrow [m-\Delta, m'+\Delta]}$ is the probability of transferring money Δ from an agent with money m to an agent with money m' per unit time. This probability, multiplied by the occupation numbers $P(m)$ and $P(m')$, gives the rate of transitions from the state $[m, m']$ to the state $[m-\Delta, m'+\Delta]$. The first term in Eq. (14) gives the depopulation rate of the state m . The second term in Eq. (14) describes the reversed process, where the occupation number $P(m)$ increases. When the two terms are equal, the direct and reversed transitions cancel each other statistically, and the probability distribution is stationary: $dP(m)/dt = 0$. This is the principle of detailed balance, or the so called equilibrium.

In physics, the fundamental microscopic equations of motion of particles obey time-reversal symmetry. This means that the probabilities of the direct and reversed processes are exactly equal:

$$f_{[m, m'] \rightarrow [m-\Delta, m'+\Delta]} = f_{[m-\Delta, m'+\Delta] \rightarrow [m, m']}. \quad (15)$$

When Eq. (14) is satisfied, the detailed balance condition for Eq. (14) reduces to the equation $P(m)P(m') = P(m-\Delta)P(m'+\Delta)$, because

the factors f cancel out. The only solution of this equation is the exponential function $P(m) = c \exp(-m/T_m)$. So the Boltzmann-Gibbs distribution is the stationary solution of the Boltzmann kinetic equation.

As previously discussed, in the case of proportional transfers (11), the time-reversal symmetry is broken, and the stationary distribution is indeed not a Boltzmann-Gibbs distribution, but a Gamma distribution. Indeed the payer pays a deterministic fraction of his money, but the receiver receives a random amount from a random agent, so their roles are not interchangeable. Therefore, the stationary distribution $P(m)$ in multiplicative models is typically not exactly exponential.

These examples show that the Boltzmann-Gibbs distribution is not valid for any conservative model. We can state that for all conservative models that have time-reversal symmetry, the stationary distribution is exponential and does not depend on the details of the model.

2.5 Statistical Mechanics of Wealth

One can argue that wealth w_i is equal to money m_i plus the other property that an agent i has. Money (paper cash, bank accounts) is generally liquid and countable. However, the other property is not immediately liquid and has to be sold first (converted into money) to be used for other purchases. In order to estimate the monetary value of property, one needs to know the price p . In the simplest model, let us consider just one type of property, say, stocks s . Then the wealth of an agent i is given by the formula

$$w_i = m_i + p s_i, \quad (16)$$

where s_i is the number of stocks owned by the agent i . It is assumed that the price p is the same for all agents.

It is reasonable to start with a model where both the total money $M = \sum_i m_i$ and the total stock $S = \sum_i s_i$ are conserved. Although M and S are conserved, the total wealth $W = \sum_i w_i$ is generally not conserved, because of price fluctuation. Here the wealth w_i of an agent i , not participating in any transactions, may change when transactions between other agents establish a new price p . More-

over, the wealth w_i of an agent i does not change after a transaction with an agent j . Indeed, in exchange for paying money Δm , agent i receives the stock $\Delta s = \Delta m/p$, so her total wealth (remains the same. In principle, the agent can instantaneously sell the stock back at the same price and recover the money paid.

Many models ([1, 10, 11]) that aim to describe the price fluctuations of p introducing a demand-supply mechanism, where the price is determined by the balance between supply and demand, demonstrated via computer simulations and proved analytically using the theory of Markov processes that the stationary distribution $P(w)$ of wealth w is given by the Gamma distribution. In general

2.5.1 Wealth growth as a stochastic process

3 Results

3.1 Basic Money Distribution

In this scenario, we consider a simple model of money distribution among agents without debt. The total money of the system is conserved, and no agent can have a negative balance. The agents engage with each other randomly but always with a fixed amount of money in each transaction.

Said so, we can assume that the money distribution will converge to the Boltzmann-Gibbs distribution as predicted by the theoretical framework. It is possible to see the results in Figure 1, where the histogram of the money distribution is plotted along with the theoretical Boltzmann-Gibbs distribution.

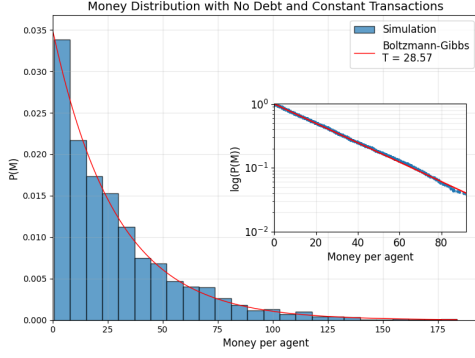


Figure 1: The histogram shows the money distribution of agents after a 10^6 number of transactions, while the red line represents the theoretical Boltzmann-Gibbs distribution. The simulation was performed with $N = 3500$, $M = 10000$, and $\Delta m = 1$. Agents in this scenario can't have debt.

Later on it will be discussed how the richer agents don't really follow Boltzmann-Gibbs distribution, but they follow a power law distribution, as observed in real economies. These agents are not in equilibrium with the rest of the system, therefore they are not part of the thermalized system.

3.2 Money Distribution with Debt

The situation now is similar but in this case the agents can have negative balances up to a maximal threshold debt (md). As it is shown in Fig. 2, after many interactions, the population seems to follow a Boltzmann-Gibbs distribution.

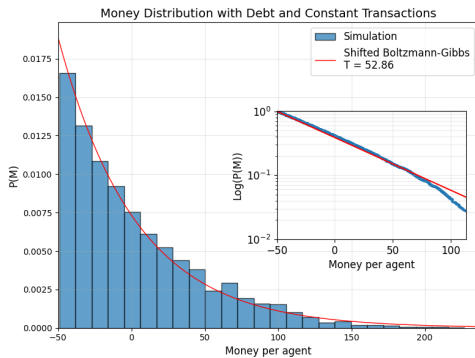


Figure 2: The histogram shows the money distribution of agents after a 10^6 number of transactions, while the red line represents the theoretical Boltzmann-Gibbs distribution. The simulation was performed with $N = 3500$, $M = 10000$, and $\Delta m = 1$. Agents in this scenario can have negative balances and the maximal debt is $md = 50$.

One way of limiting the total debt in the economy, as we explained in section 2.3.2, is the so-

called required reserve ratio r [12]. The value of r was set to $r = 0.618$ in order to have the two branches of the double-sided Boltzmann-Gibbs distribution meet at the origin. The computation of this value is shown in Appendix A.

The results are shown in Fig. 3, where the histogram of the money distribution is plotted along with the theoretical double-sided Boltzmann-Gibbs distribution.

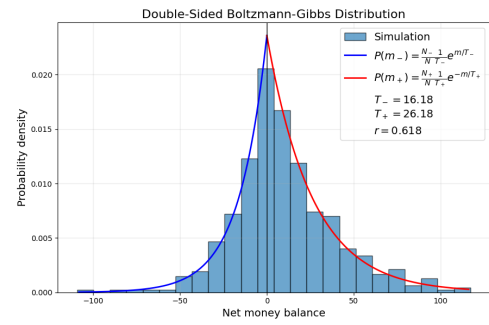


Figure 3: The histogram shows the money distribution of agents after 10^7 number of transactions, while the red and blue lines represent the theoretical double-sided Boltzmann-Gibbs distribution. The simulation was performed with $N = 500$, $M = 5000$, and $\Delta m = 1$. The reverse ratio is set to $r = 0.618$ so that the two branches of the distribution meet at the origin.

3.3 Money Distribution with Saving Propensity

Let's now consider a scenario where agents have a saving propensity γ . Each agent can exchange only a fraction of their money, which is determined by the saving propensity. The results of the simulation are shown in Fig. 4, where the histogram of the money distribution is plotted along with the theoretical Gamma distribution.

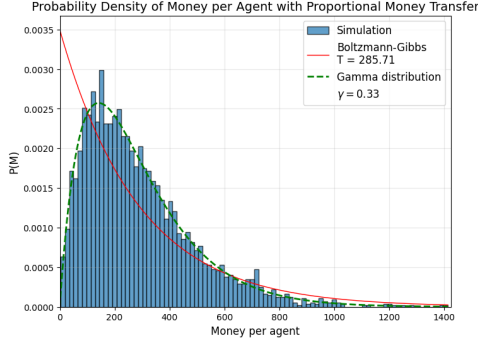


Figure 4: The histogram shows the money distribution of agents after 10^6 number of transactions, while the green line represents the theoretical Gamma distribution and the red line represents the Boltzmann-Gibbs distribution with such temperature. The simulation was performed with $N = 3500$, $M = 10000$, and $\gamma = 1/3$. Agents in this scenario can exchange only a fraction of their money, determined by the saving propensity γ .

3.4 Power Law Distribution

4 Conclusion

In the absence of detailed knowledge of real microscopic dynamics of economic exchanges, the semi-universal Boltzmann-Gibbs distribution is a natural starting point.

4.1 Summary

[Conclusion content to be filled...]

A Mathematical Details

A.1 Boltzmann-Gibbs Distribution

[Mathematical details to be filled...]

A.2 Reverse Ratio choice of value r

From section 2.3.2, we can see that the choice of r is crucial for the model to work properly. The choice of r is based on the assumption that the total debt in the system is limited by the required reserve ratio. To have matching populations at $m = 0$, the condition $P(m_+) = P(m_-)$ reduces to

$$\frac{N_+}{T_+} = \frac{N_-}{T_-},$$

with

$$T_+ = \frac{M_+}{N_+}, \quad T_- = \frac{M_-}{N_-}.$$

Using the definitions of the two sectors,

$$N_+ = rN, \quad N_- = (1-r)N,$$

$$M_+ = \frac{M_0}{r}, \quad M_- = \frac{M_0(1-r)}{r},$$

we obtain

$$T_+ = \frac{M_0}{r^2 N}, \quad T_- = \frac{M_0}{r N}.$$

Substituting these expressions into the matching condition yields

$$r \cdot \frac{r^2 N}{M_0} = (1-r) \cdot \frac{r N}{M_0},$$

which simplifies to

$$r^2 + r - 1 = 0.$$

Therefore the reverse ratio that makes the two distributions meet at the origin is

$$r = \frac{\sqrt{5} - 1}{2} \approx 0.618.$$

This is the value used in the simulations to compare the positive and negative branches of the double-sided Boltzmann-Gibbs distribution.

B Code Snippets

B.1 Core Simulation

[Code snippets to be filled...]

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